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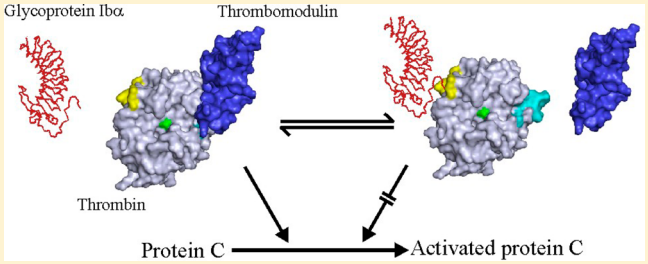
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Exosite 2-Directed Ligands Attenuate Protein C Activation by the Thrombin−Thrombomodulin Complex

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ABSTRACT: Thrombin activity, inhibition, and localization are regulated by two exosites that flank the active site. Substrates, cofactors, and inhibitors bind to exosite 1 to promote active site access, whereas exosite 2 interactions hold thrombin on cells, platelets, and proteins. The exosites also serve allosteric roles, whereby ligand binding alters thrombin activity. Previously, we showed that ligands that bind exosite 2 attenuate the exosite 1-mediated interaction of thrombin with fibrin, demonstrating allosteric connection between the exosites. To determine the functional consequences of these inter-exosite interactions, we examined the effect of exosite 2 ligands on thrombin's interaction with thrombomodulin, a key cofactor that binds exosite 1 and redirects thrombin activity to the anticoagulant protein C pathway. Exosite 2-directed ligands, which included the HD22 aptamer, glycoprotein 1bα-derived peptide, and fibrinogen γ'-chain peptide, reduced the level of exosite 1-mediated thrombin binding to the thrombomodulin peptide consisting of the fourth, fifth, and sixth epidermal-like growth factor-like domains, decreasing affinity by >10-fold, and attenuated thrombomodulin-dependent activation of protein C by 60−80%. The ligands had similar effects on thrombin-mediated protein C activation with intact soluble thrombomodulin and with thrombomodulin on the surface of cultured endothelial cells. Their activity was exosite 2-specific because it was attenuated when RA-thrombin, a variant lacking exosite 2, was used in place of thrombin. These results indicate that additional reactions mediated by exosite 1 are amenable to regulation by exosite 2 ligation, providing further evidence of inter-exosite allosteric regulation of thrombin activity.



Thrombin is the central mediator of hemostasis. It acts primarily as a procoagulant enzyme by activating platelets and by converting soluble fibrinogen into insoluble fibrin, the structural backbone of blood clots. Furthermore, thrombin amplifies its own generation by activating factors V, VIII, and XI, key procoagulant factors. When thrombin is bound to thrombomodulin, the specificity of thrombin is changed from a procoagulant to an anticoagulant and antifibrinolytic enzyme. The anticoagulant effect of the thrombin−thrombomodulin complex reflects its capacity to convert protein C to activated protein C, which inhibits thrombin generation by inactivating activated factor V and VIII (factor Va and factor VIIIa, respectively).[1] The antifibrinolytic effect of the thrombin−thrombomodulin complex reflects activation of thrombin-activatable fibrinolysis inhibitor, which attenuates clot lysis by removing carboxy-terminal Lys and Arg residues, thereby reducing the capacity of degrading fibrin to promote plasminogen activation.[2] Therefore, regulation of these seemingly antagonistic roles of thrombin is critical for hemostasis.

Two important domains on thrombin, known as exosites, regulate its diverse functions. The exosites are positively charged domains that lie on opposite sides of the active site cleft of thrombin.[3,4] Many thrombin substrates, such as fibrinogen, factor V, factor VIII, and protease-activated receptors (PARs), as well as thrombin inhibitors, such as heparin cofactor II and hirudin, bind to exosite 1. All of these interactions precede subsequent engagement at the active site.[4] Thrombin cofactors, such as thrombomodulin, also bind to exosite 1 to modulate the anticoagulant and antifibrinolytic properties of thrombin.[5] Although thrombomodulin on the endothelial cell surface possesses varying amounts of exosite 2 binding chondroitin sulfate, the exosite 1-dependent binding of thrombin to the fourth, fifth, and sixth epidermal growth factor domains is essential for the cofactor activity of thrombomodulin.[6,7] Other exosite 2-mediated interactions are predominantly involved in the localization or regulation of thrombin. For example, platelet glycoprotein Ibα (GpIbα) binds exosite 2 and localizes thrombin on the platelet surface, which facilitates its activation of PARs.[8] Likewise, the extended γ-chain of a variant form of fibrinogen, termed γ'-fibrinogen, binds exosite 2 and mediates interaction of thrombin with an affinity that is higher than that with the bulk γ_A-fibrinogen and greater protection from inhibition by antithrombin and heparin cofactor II.[9]

Received: March 17, 2017
Revised: May 5, 2017
Published: May 26, 2017

ACS Publications © 2017 American Chemical Society

DOI: 10.1021/acs.biochem.7b00250
Biochemistry 2017, 56, 3119−3128

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Regulation of thrombin activity is not limited to ligation of substrates, cofactors, and inhibitors. The exosites also regulate thrombin activity via allosteric control. Thus, binding of thrombomodulin to exosite 1 or binding of prothrombin fragment 2 to exosite 2 exerts functional effects on the active site.[10−14] Furthermore, these allosteric effects have been shown to extend to the other exosite, indicative of the global response of thrombin to ligand binding.[14−23] Thus, binding of peptide or aptamer ligands to one exosite on thrombin exerts functional effects at the other exosite. Although disputed with small synthetic ligands,[24,25] inter-exosite allostery has been shown to occur with physiological ligands such as fibrin, where thrombin binding via exosite 1 is attenuated by exosite 2 ligands.[11] Thus, to examine the functional consequences of this interaction, we investigated whether exosite 2-specific ligands influence binding of thrombin to thrombomodulin and protein C activation by the thrombin−thrombomodulin complex. A thrombomodulin-derived peptide that possesses only the fourth, fifth, and sixth epidermal-like growth factor-like domains (TM456) was used in these studies because it binds thrombin exclusively via exosite 1.[22] To confirm that the results with TM456 reflect those with full-length thrombomodulin, we also examined the effect of the exosite 2-directed ligands on thrombin-mediated protein C activation in the presence of Solulin, a recombinant analogue of the extracellular portion of thrombomodulin lacking the chondroitin sulfate moiety,[26] and by thrombomodulin on the surface of cultured human umbilical vein endothelial cells (HUVEC). We demonstrate that γ′-peptide, an analogue of the carboxy terminus of the γ′-chain of fibrinogen, a GpIbα peptide analogue, and the aptamer HD22, all of which have been shown by us and by others to bind thrombin solely via exosite 2,[8,9,27−31] disrupt the thrombin−TM456 interaction and, by so doing, attenuate protein C activation by the thrombin−TM456 and thrombin−Solulin complexes and by thrombin on the surface of HUVEC. These results demonstrate that binding of a ligand to exosite 2 has the potential to modulate the exosite 1-dependent interaction of thrombin with thrombomodulin, thereby attenuating thrombin's capacity to activate protein C.

EXPERIMENTAL PROCEDURES

Materials. Human protein C and thrombin were from Haematologic Technologies Inc. (Essex Junction, VT) and Enzyme Research Laboratories (South Bend, IN), respectively. Thrombin was dialyzed against 20 mM HEPES (pH 7.4) and 150 mM NaCl (HBS) to eliminate citrate. RA-thrombin, a mutant thrombin with Arg 93, Arg 97, and Arg 101 replaced with Ala, was a generous gift from C. Esmon and was prepared as described previously.[27,32] Thrombin was inactivated with D-Phe-Pro-Arg chloromethyl ketone (FPR; EMD Millipore) as described previously.[33] Hirudin was from Dade-Behring (Marburg, Germany). Oligonucleotides HD1 (5′-GGTTGGTGTGGTTGG-3′), an exosite 1-specific DNA aptamer, HD22 (5′-AGTCCGTGGTAGGGCAGGTTGGGGTGACT-3′), an exosite 2-specific DNA aptamer, and HD23 (5′-AGTCCGTAATAAAGCAGGTTAAAATGACT-3′), an inactive variant of HD22, were synthesized by the Molecular Biology and Biotechnology Institute at McMaster University (Hamilton, ON).[27,34] Aptamers were dissolved in aptamer buffer consisting of 20 mM Tris and 150 mM NaCl (pH 7.4) (TBS), containing 2 mM $CaCl_2$, 5 mM KCl, 1 mM $MgCl_2$, and 0.1% polyethylene glycol.[35] Before use, the aptamer preparations were heated to 95 °C for 10 min and then cooled on ice for an additional 10 min to permit renaturation.[36] γ′-Peptide [H-Val-Arg-Pro-Glu-His-Pro-Ala-Glu-Thr-Glu-Tyr(H_2PO_3)-Asp-Ser-Leu-Tyr-(H_2PO_3)-Pro-Glu-Asp-Asp-Leu], a Tyr-phosphorylated 20-amino acid analogue of the carboxy-terminal portion of the γ′-chain of fibrinogen,[37] was from Bachem AG (San Diego, CA). GpIbα269PPP, a Tyr-phosphorylated peptide analogue of residues 269−286 of GpIbα [H-Asp-Glu-Gly-Asp-Thr-Asp-Leu-Tyr(PO_3H_2)-Asp-Tyr(PO_3H_2)-Tyr(PO_3H_2)-Pro-Glu-Glu-Asp-Thr-Glu-Gly],[8] and GpIbα269scr, a nonphosphorylated, scrambled version of this peptide (H-Asp-Gly-Glu-Thr-Asp-Leu-Asp-Tyr-Asp-Tyr-Glu-Tyr-Glu-Pro-Asp-Thr-Glu-Gly), were prepared by Mimotopes (Clayton, Australia). TM456 was a generous gift from T. Mather of the University of Oklahoma (Oklahoma City, OK).[38] Solulin, a recombinant analogue of the extracellular portion of TM lacking the chondroitin sulfate moiety,[39] was generously provided by K. E. Peterson (Paion, Aachen, Germany). Chromogenic substrates pyroGlu-Pro-Arg-*p*-nitroanilide (S-2366) and tosyl-Gly-Pro-Arg-4-nitroanilide acetate (Chromozym TH) were purchased from Chromogenix and Roche Applied Science, respectively. Chondroitinase ABC was obtained from Sigma-Aldrich.

Surface Plasmon Resonance (SPR) Analysis of the Interaction of Thrombin with Immobilized TM456. The interaction of thrombin, FPR-thrombin, or RA-thrombin with immobilized TM456 was quantified using SPR on a Biacore T200 (GE Healthcare). TM456 was biotinylated with biotinaminohexanoic acid-3-sulfoester and adsorbed to a flow cell of a CM5 biosensor chip containing immobilized streptavidin to 150 response units (RU) as described previously.[11] Samples for SPR were prepared in HBS containing 0.05% Tween 20 and 0.1 mM $CaCl_2$. Increasing concentrations (0−50 nM) of active thrombin, FPR-thrombin, or RA-thrombin were injected into flow cells containing immobilized TM456 at 23 °C and a flow rate of 100 μL/min to quantify binding. RU values obtained from the sensorgrams were plotted versus thrombin concentration, and the data were analyzed by nonlinear regression of a rectangular hyperbola to obtain K_d.[40] Flow cells were regenerated with 1.5 M NaCl between runs.

In competition experiments, thrombin or RA-thrombin was injected in the presence of increasing concentrations of ligands to determine their effect on the binding of thrombin to TM456. In these experiments, 10 nM thrombin or 25 nM RA-thrombin was injected in the presence of 0−1000 nM HD1, 0−600 nM HD22, 0−600 nM HD23, 0−100 μM γ′-peptide, or 0−10 μM GpIbα269PPP. The volume fraction of aptamer buffer was kept constant in all aptamer-containing samples. BIAcore experiments with GpIbα269PPP were performed in the presence of 0.1 mg/mL BSA. RU values were determined for each condition using instrument software and plotted versus the concentration of the ligand. EC_{50} values for disruption of binding of thrombin or RA-thrombin to TM456 were determined by rectangular hyperbola analyses.

To determine the affinity of thrombin for TM456 in the presence of exosite ligands, 0−50 nM thrombin was injected in the presence of 1000 nM HD1, 300 nM HD22, 300 nM HD23, 100 μM γ′-peptide, or 10 μM GpIbα269PPP. For RA-thrombin, 0−50 nM was injected in the presence of 600 nM HD22 or 100 μM γ′-peptide. Association and dissociation rate constants were calculated by the instrument software by fitting the binding data to two-state reaction and steady-state affinity models to determine the $K_{d,app}$.

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Biochemistry 2017, 56, 3119−3128

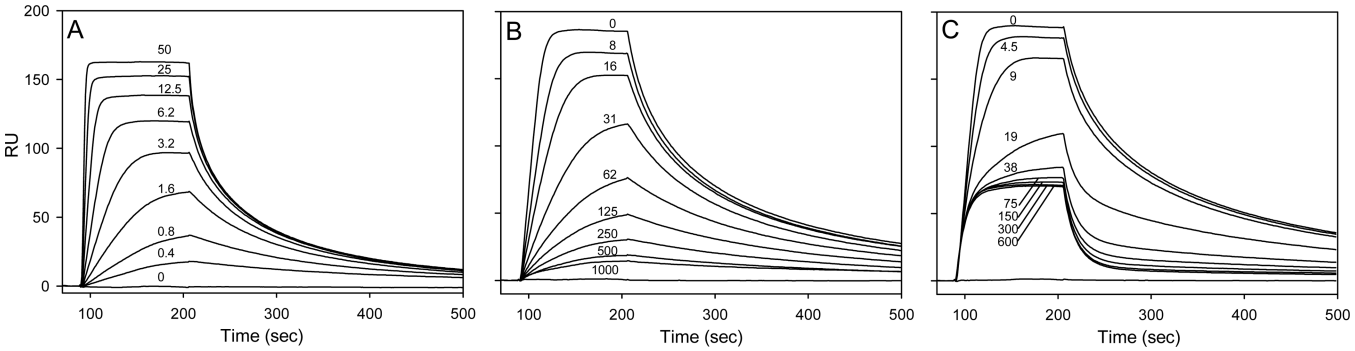


Figure 1. Sensorgrams of binding of thrombin to TM456 in the absence or presence of exosite ligands. (A) Thrombin at concentrations ranging from 0 to 50 nM was injected from 90 to 210 s into a flow cell containing TM456 immobilized on a CM5 chip. Buffer was subsequently injected for 300 s to monitor dissociation. Response units (RU) are corrected for flow over a blank cell and are plotted vs time. Thrombin concentrations (nanomolar) are indicated. (B and C) Thrombin (10 nM) was injected into a flow cell containing immobilized TM456 in the presence of increasing concentrations of HD1 and HD22, respectively, followed by buffer. Aptamer concentrations (nanomolar) are indicated adjacent to lines.

Effect of Exosite-Directed Ligands on Protein C Activation. The capacity of HD1, HD22, HD23, γ′-peptide, GpIbα269PPP, and GpIbα269scr to attenuate the activation of protein C by the thrombin−TM456 complex was investigated using a discontinuous assay at 23 °C. Reaction mixtures containing 200 nM protein C, 20 nM thrombin, and 20 nM TM456, with varying concentrations of HD1 (0−1000 nM), HD22 (0−1000 nM), HD23 (0−1200 nM), γ′-peptide (0−80 μM), GpIbα269PPP (0−30 μM), or GpIbα269scr (0−30 μM), were made in TBS containing 0.1 mM $CaCl_2$. This concentration of $CaCl_2$ was chosen because thrombin-mediated protein C activation in the presence of thrombomodulin lacking chondroitin sulfate is maximal at 0.1 mM $CaCl_2$ and is attenuated or inhibited with higher $CaCl_2$ concentrations.[22,32] Reaction mixtures were incubated for 15 min prior to the addition of 200 nM hirudin to inhibit thrombin. After incubation for 5 min, 400 μM S-2366 was added and activated protein C generation was quantified by monitoring substrate hydrolysis at 405 nm in kinetic mode using a SpectraMax 340PC384 plate reader (Molecular Devices, Sunnyvale, CA). Rates of hydrolysis were normalized relative to that obtained in the absence of exosite-directed ligands and plotted against the concentration of the exosite-directed ligand. Nonlinear regression of the data using a rectangular hyperbolic equation was performed with Table Curve (Jandel Scientific, Systat Software Inc., San Jose, CA) to yield EC_{50} values that represent a half-maximal effect.

Experiments were repeated using 40 nM RA-thrombin in place of thrombin and 30 nM TM456, or using 20 nM thrombin or 40 nM RA-thrombin and 20 nM Solulin in place of TM456. Titrations of HD22, γ′-peptide, and GpIbα269PPP were performed in the absence of TM456 or Solulin and $CaCl_2$ to account for effects independent of exosite 1−TM456 interactions. Calcium was omitted from these controls because it inhibits protein C activation by thrombin in the absence of thrombomodulin.[38] Thrombin and protein C were incubated for 2 h in the absence of TM456 before activated protein C chromogenic activity was determined as described above.

Effect of Exosite Ligands on the Apparent Affinity of Thrombin for TM456. The apparent affinity ($K_{d,app}$) of TM456 for thrombin was determined in the absence or presence of HD1, HD22, γ′-peptide, and GpIbα269PPP. Activation of 200 nM protein C by 20 nM thrombin in the presence 0−120 nM TM456 was performed in TBS buffer containing 0.1 mM $CaCl_2$ in the absence or presence 600 nM HD1, 100 nM HD22, 80 μM γ′-peptide, or 30 μM GpIbα269PPP. Reaction mixtures were incubated for 10 min prior to being quenched with 200 nM hirudin. Activated protein C activity was determined as described above. Nonlinear regression analysis of a rectangular hyperbola was performed using Table Curve to yield values for $K_{d,app}$ and saturating chromogenic activity.

Effect of Exosite Ligands on Protein C Activation on the Surface of HUVEC. HUVEC (Lonza, Walkersville, MD) in endothelial cell growth medium-2 (EGM-2 MV Bullet Kit, Lonza) were cultured in 5% CO_2 at 37 °C in 2% gelatin-coated T-75 flasks or in the wells of 24-well plates (BD Falcon, Bedford, MA) as described previously.[41] Cells were used on passages 2−4 when they were 80−90% confluent. Immediately before use, cells were rinsed with Hank's balanced salt solution containing 3 mM $CaCl_2$, 0.6 mM $MgCl_2$, and 0.5% human serum albumin (wash buffer). In duplicate wells, HUVEC were incubated with protein C (1 μM) and 10 nM thrombin or 20 nM RA-thrombin for 30 min at 25 °C in the presence of 0−3 μM HD22, HD1, or HD23 or 0−1 mM γ′-peptide in wash buffer. After 100 μL aliquots of supernatant had been transferred to wells of a 96-well plate containing 100 μL of 100 nM hirudin and 1 mM S-2366, reactions were monitored at 405 nm in a SpectraMax Plus plate reader, and initial rates of S-2366 hydrolysis were determined from the slopes of plots of absorbance versus time. Rates of substrate cleavage were converted to activated protein C concentrations, using the specific chromogenic activity determined separately. To examine the role of the chondroitin sulfate moiety of thrombomodulin, experiments were repeated with HUVEC that were pretreated with chondroitinase ABC. Cells grown as described above were incubated with 0.2 unit/mL chondroitinase ABC for 45 min at 37 °C, as described previously,[42] prior to washing and measuring protein C activation as outlined above.

Statistical Analyses. Mean and standard deviation values were obtained from experiments performed in triplicate.

RESULTS

Effect of Exosite Ligands on Binding of Thrombin to Immobilized TM456 Measured by SPR. SPR was used to examine the effect of exosite ligands on binding of active thrombin to TM456. Injection of increasing concentrations of thrombin over a flow cell containing immobilized TM456 was used to determine the dissociation constant (K_d) for binding.

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DOI: 10.1021/acs.biochem.7b00250
Biochemistry 2017, 56, 3119−3128

Under these conditions, thrombin bound immobilized TM456 in a saturable and reversible manner with a K_d value of 1.6 ± 0.2 nM (Figure 1A). This is comparable with the value of 1.9 nM reported previously,[43] slightly higher than the K_d value of 0.6 nM revealed using SPR,[39] but lower than the value of 6.2 nM obtained by fluorescence.[12] Subsequent studies examined the influence of exosite 1- and 2-directed ligands on binding of thrombin to immobilized TM456 (Figure 1B,C). In these experiments, a fixed concentration of thrombin was injected in the presence of increasing concentrations of aptamer HD1, HD22, or HD23 or γ′-peptide. As expected, because of its specificity and high affinity for exosite 1, HD1 prevents thrombin from binding to TM456, with an EC_{50} of 49.4 ± 4.1 nM (Table 1).[44] To test for the involvement of exosite 2, HD22 and γ′-peptide were used. Both exosite 2 ligands caused partial (~50%) but saturable reductions in the level of binding of thrombin to TM456 (Figures 2 and 3). As expected because of its higher affinity for thrombin, HD22 was more potent than γ′-peptide at attenuating binding of thrombin to TM456, with EC_{50} values of 14.4 ± 1.0 nM and 45.5 ± 8.4 μM, respectively.[11] In contrast, HD23, an aptamer that does not bind to thrombin, had no effect on the thrombin−TM456 interaction (Figure 2). A third ligand for exosite 2 binding, a peptide fragment of GpIbα containing phosphorylated Tyr residues (GpIbα269PPP), was also examined.[8] GpIbα269PPP reduced the level of thrombin binding by 50% with an EC_{50} of 49.1 ± 5.6 μM (not shown). Our findings with HD22 and γ′-peptide are consistent with those of previous studies that showed that exosite 2 ligands reduced the level of binding of thrombin to fibrin and provide additional evidence of inter-exosite allosteric effects.[11]

Table 1. Inhibition of Binding of α-Thrombin to Immobilized TM456[a]

ligand	EC_{50}
HD1	49.4 ± 4.1 nM
HD22	14.4 ± 1.0 nM
γ′-peptide	45.5 ± 8.4 μM
GpIbα269PPP	49.1 ± 5.6 μM

[a]Binding of thrombin to immobilized TM456 was assessed by SPR in the presence of increasing concentrations of exosite ligands as shown in Figures 2 and 3. Values for EC_{50} for reduction of the level of thrombin binding were determined by nonlinear regression. Values represent the means ± SD of three determinations.

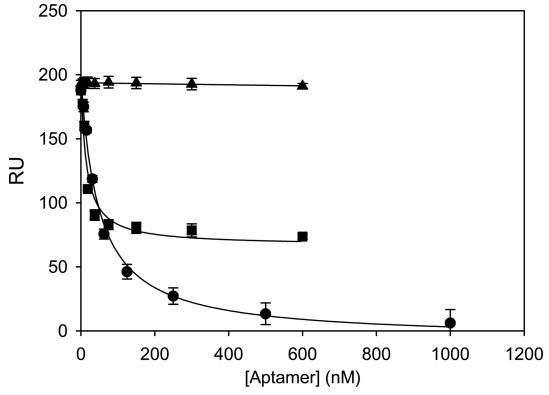


Figure 2. Effect of exosite-directed ligands on the binding of thrombin to TM456. TM456 was immobilized on a CM5 chip to 150 RU in HBS containing 0.05% Tween 20 and 0.1 mM $CaCl_2$. Thrombin (10 nM) was injected at a flow rate of 100 μL/min in the presence of 0−1000 nM HD1 (●), 0−600 nM HD22 (■), or 0−600 nM HD23 (▲). RU values were determined from the sensorgrams. Data were plotted vs ligand concentration and analyzed by rectangular hyperbola to obtain EC_{50} values (lines). Symbols represent the means ± SD of three experiments.

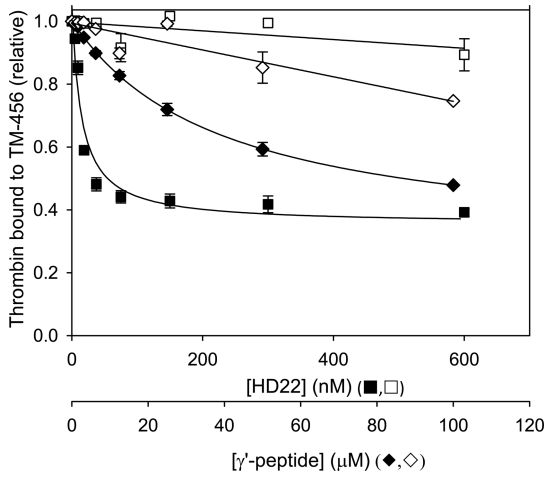


Figure 3. Effect of exosite 2-directed ligands on the binding of thrombin or RA-thrombin to TM456. Thrombin at 10 nM (filled symbols) or RA-thrombin at 25 nM (empty symbols) in HBS containing 0.1 mM $CaCl_2$ and 0.05% Tween 20 was injected into flow cells containing immobilized TM456 in the presence of 0−100 μM γ′-peptide (diamonds) or 0−600 nM HD22 (squares). The level of thrombin bound relative to that determined in the absence of ligand was determined and plotted vs ligand concentration. Linear regression was used to determine EC_{50} values (lines). Symbols represent the means ± SD of three experiments.

The specificity of the effects on binding was examined using RA-thrombin, an exosite 2 variant with affinity for heparin and γ′-peptide >20-fold lower than that for thrombin.[27,32] HD22 and γ′-peptide had little effect on binding of RA-thrombin to TM456, demonstrating potency >7-fold lower than that of thrombin (Figure 3). These results indicate that these two ligands bind thrombin via exosite 2, thereby confirming that the responses with thrombin represent inter-exosite communication.

The involvement of the active site was investigated with D-Phe-Pro-Arg chloromethyl ketone (FPR)-inactivated thrombin (not shown). FPR-thrombin bound immobilized TM456 with an affinity (K_d value of 7.5 ± 0.2 nM) 5-fold lower than that of active thrombin, likely because of reciprocal allosteric communication between TM binding at exosite 1 and the active site.[13,45,46] HD22 displaced FPR-thrombin from TM456 with an EC_{50} of 62.6 ± 1.8 nM, demonstrating potency 5-fold lower than with active thrombin. The lower potency could reflect similar FPR perturbation of communication between the active site and exosite 2.[20] These results demonstrate that active site occupation does not negate inter-exosite communication.

Given the reduction in the level of binding, the affinities of thrombin for TM456 were determined in the presence of saturating concentrations of HD1, HD22, γ′-peptide, and GpIbα269PPP (not shown). In the presence of 1000 nM HD1, the affinity of thrombin for TM456 was reduced 16-fold from 1.6 ± 0.2 to 26.7 ± 2.4 nM, whereas in the presence of 300 nM HD22, the affinity of thrombin for TM456 was reduced 13-fold to 20.4 ± 1.0 nM. Furthermore, in the presence of 100 μM γ′-

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DOI: 10.1021/acs.biochem.7b00250
Biochemistry 2017, 56, 3119−3128

peptide or 10 μM GpIbα269PPP, the affinity of thrombin for TM456 was reduced 5- or 26-fold to 7.7 ± 0.2 or 42.3 ± 2.1 nM, respectively. HD23 had no effect on the binding of thrombin to TM456. With RA-thrombin, HD22 had no effect on its binding to TM456, whereas γ'-peptide reduced the affinity by 2-fold. These results confirm that exosite 2 ligands alter the affinity of thrombin for TM456, an interaction solely mediated by exosite 1.

Effect of Exosite-Directed Ligands on Protein C Activation. We next investigated the ability of the exosite-directed ligands to attenuate protein C activation by the thrombin−TM456 complex. For comparison purposes, results are shown as relative rates of protein C activation compared with control in the absence of exosite ligand. In control experiments, the rate constant of protein C activation by thrombin was 3.1 μM^{-1} s^{-1} in the presence of saturating concentrations of TM456, comparable with values reported in the literature.[12] As a positive control, HD1 fully inhibited activation of protein C with a half-maximal effect (EC_{50}) at 40.7 ± 2.9 nM, consistent with an exosite 1-mediated interaction (Figure 4A and Table 2). HD22 and γ'-peptide attenuated protein C activation by the thrombin−TM456 complex by >80% with EC_{50} values of 27 ± 3.1 nM and 13.9 ± 1.3 μM, respectively, whereas GpIbα269PPP attenuated activation by >60% with an EC_{50} of 4.5 ± 0.1 μM. Neither HD23 nor GpIbα269scr had saturable effects (Figure 4A,B).

Control experiments were repeated at 200 nM TM456 where thrombin was fully saturated with the cofactor (not shown). Under these conditions, the EC_{50} for HD22 was 35.6 ± 3.9 nM, a value comparable with the results obtained at 20 nM TM456. These data indicate that the extent of thrombin saturation of TM456 does not influence the allosteric effect.

We next investigated whether soluble thrombomodulin would elicit the same response as TM456. This was tested using Solulin, a recombinant soluble thrombomodulin analogue containing all six epidermal-like growth factor domains.[26] Solulin was comparable with TM456 in terms of enhancing protein C activation by thrombin (not shown). Furthermore, promotion of protein C activation by TM456 and Solulin was attenuated to similar extents by HD1 and HD22 (Figure 5), and γ'-peptide and GpIbα269PPP (not shown), indicating that inter-exosite allostery is independent of the form of thrombomodulin that is used.

To verify that the inhibitory effects were a result of exosite 2-specific interactions, and not due to nonspecific binding of ligands to exosite 1, we repeated these experiments using RA-thrombin (Figure 4A,B). As with native thrombin, HD1 abrogated TM456-mediated protein C activation by RA-thrombin with an EC_{50} value of 25.7 ± 1.2 nM (Table 2). In contrast, HD22, γ'-peptide, and GpIbα269PPP had no effect on protein C activation by RA-thrombin. These findings suggest that the observed inhibitory effects of these ligands on the thrombin−TM456 interaction were due to ligand binding specifically at exosite 2.

Next, we activated protein C in the absence of TM456 to examine the effects of the exosite 2 ligands on the active site alone. Thrombin is a weak activator of protein C in the absence of thrombomodulin and reacts solely via the active site. These experiments were performed in the absence of Ca^{2+}, which impairs protein C activation by thrombin alone.[38] All three exosite 2-directed ligands exhibited weak attenuation of protein C activation in the absence of TM456. The magnitude of attenuation was less than half that observed when TM456 was

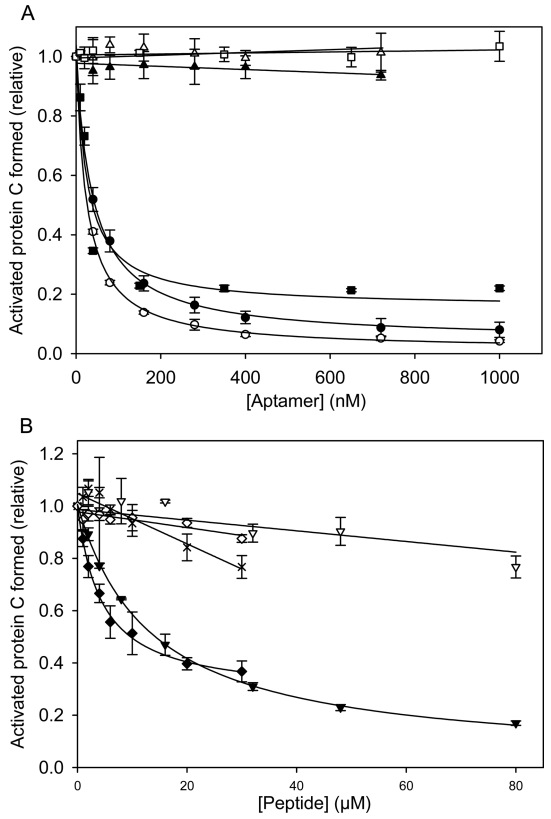


Figure 4. Effect of exosite-directed ligands on TM456-dependent protein C activation by thrombin or RA-thrombin. Protein C (200 nM) and 20 nM TM456 in TBS containing 0.1 mM $CaCl_2$ were incubated with 20 nM thrombin (filled symbols) or 40 nM RA-thrombin (empty symbols) for 15 min in the presence of aptamer (A) 0−1000 nM HD1 (circles), 0−1000 nM HD22 (squares), or 0−700 nM HD23 (triangles) or peptide (B) 0−80 μM γ'-peptide (inverted triangles) or 0−30 μM GpIbα269PPP (diamonds) prior to addition of hirudin to a final concentration of 200 nM. Control GpIbα269scr peptide (0−30 μM) was examined only with thrombin (×). After incubation for 5 min, 400 μM S-2366 was added and activated protein C generation was quantified by monitoring substrate hydrolysis at 405 nm and normalized relative to that determined in the absence of ligand. Data were analyzed by nonlinear regression of a rectangular hyperbola to determine EC_{50} values (lines). Symbols represent the means ± SD of three experiments.

present (Figure 6). These results indicate that the exosite 2 ligands also exert allosteric effects at the active site. However, the decrease in the level of protein C activation by exosite 2 ligands is greater when TM456 is present, indicating the potential for modulation under physiological conditions.

Finally, we examined the effect of the exosite ligands on thrombin-mediated protein C activation on the surface of HUVEC to determine whether they elicited the same response with cell surface-associated thrombomodulin. HUVEC were incubated with thrombin, protein C and increasing concentrations of HD1, HD22, HD23, or γ'-peptide for 30 min. Activated protein C activity was determined from the culture supernatants using a chromogenic assay. HD22 and γ'-peptide attenuated protein C activation by 40 and 100%, respectively (Figure 7). HD1 also inhibited activation by 100%, whereas HD23 had no effect. EC_{50} values were 12 and 18 nM for HD1 and HD22, respectively, whereas the EC_{50} value was 120 μM for γ'-peptide. As expected, the level of activation of protein C on HUVEC by RA-thrombin was reduced by HD1, but not by

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Biochemistry 2017, 56, 3119−3128

Table 2. Effect of Exosite Ligands on Protein C Activation by Thrombin in the Presence of TM456[a]

	EC_{50}			
thrombin	HD1	HD22	γ′-peptide	GpIbα269PPP
α-thrombin	40.7 ± 2.9 nM	27 ± 3.1 nM	13.9 ± 1.3 μM	4.5 ± 0.1 μM
RA-thrombin	25.7 ± 1.2 nM	>1000 nM	>80 μM	>30 μM

[a]Protein C activation by α-thrombin or RA-thrombin was assessed in the presence of TM456 and increasing concentrations of exosite ligands. EC_{50} values for inhibition of activation were determined by nonlinear regression. Values represent the means ± SD of three determinations.

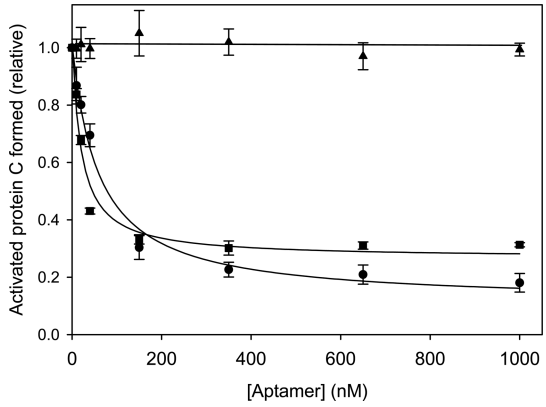


Figure 5. Effect of aptamers on protein C activation by thrombin in the presence of Solulin. Protein C (200 nM) was activated by 20 nM thrombin in TBS containing 0.1 mM $CaCl_2$ for 10 min in the presence of 20 nM Solulin and 0−3000 nM HD1 (●), HD22 (■), or HD23 (▲). After incubation with 200 nM hirudin for 5 min, activated protein C activity was detected with 400 μM S-2366. Values are normalized with respect to that determined in the absence of ligand. Data were analyzed by nonlinear regression of a rectangular hyperbola to determine EC_{50} values (lines). Symbols represent the means ± SD of three experiments.

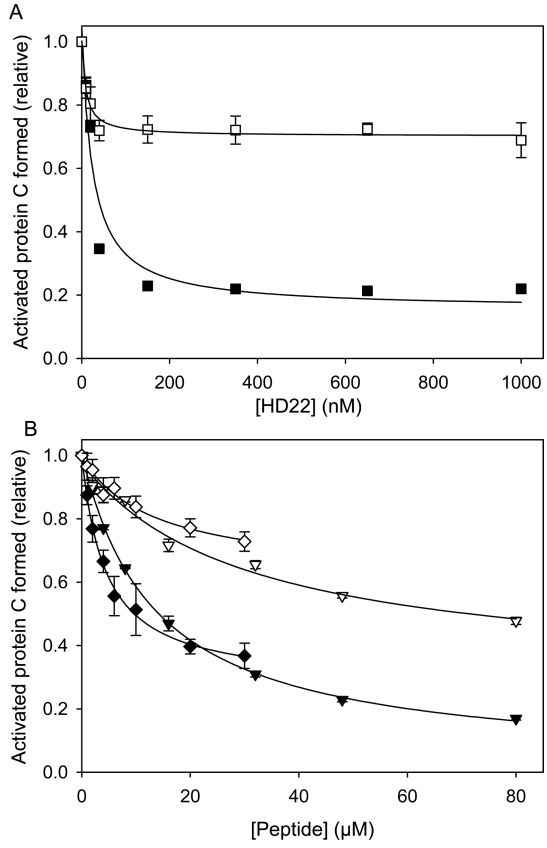


Figure 6. Effect of HD22, γ′-peptide, and GP1bα peptide on protein C activation in the absence or presence of TM456. Protein C (200 nM) was activated by 20 nM thrombin in the absence of TM456 and $CaCl_2$ (empty symbols) and with 0−1000 nM HD22 (squares, panel A), 0−80 μM γ′-peptide (inverted triangles, panel B), or 0−30 μM GpIbα269PPP (diamonds, panel B). Comparative data for activation in the presence of TM456 and $CaCl_2$ for HD22, γ′-peptide, and GpIbα269PPP from Figure 4 are shown for reference (■, ▼, and ◆, respectively). Rates of activation were determined by a chromogenic assay for activated protein C and normalized with respect to that determined in the absence of ligand. Data were analyzed by nonlinear regression of a rectangular hyperbola to determine EC_{50} values (lines). Symbols represent the means ± SD of three experiments.

HD22 or HD23 (Figure 7), confirming the specificity of HD22 in the binding and functional assays outlined above. Activation of protein C on HUVEC was unaffected by preincubation with chondroitinase ABC (not shown), indicating that the chondroitin sulfate moiety of thrombomodulin has little influence, a finding that is consistent with the observation that chondroitin sulfate contributes <20% to the activation of protein C on HUVEC.[7] Taken together, the results indicate that exosite 2 ligands modulate protein C activation whether mediated by TM456, soluble thrombomodulin, or thrombomodulin on HUVEC.

Effect of Exosite Ligands on the Thrombin−TM456 Interaction Determined by Protein C Activation. The effect of exosite 2 ligands on exosite 1 function was quantified by assessing protein C activation as a measure of the apparent affinity of TM456 for thrombin in the presence of saturating levels of competing ligands. The $K_{d,app}$ of TM456 for promotion of protein C activation by thrombin was 11 ± 0.9 nM (Figure 8 and Table 3). While this value is higher than that obtained using SPR, it is comparable with the $K_{d,app}$ of 6.2 nM reported for TM456.[12] In the presence of 600 nM HD1, the $K_{d,app}$ was >10-fold higher (117 ± 25.8 nM) than that determined in the absence of HD1. This is consistent with competitive binding of HD1 and TM456 at exosite 1. HD22 and γ′-peptide both decreased TM456 affinity by 2−3-fold [$K_{d,app}$ values of 22.3 ± 3.7 and 26.0 ± 1.6 nM, respectively (Table 3)] and decreased the rate of activation by 3-fold.

GpIbα269PPP increased the $K_{d,app}$ value by 2-fold (20.8 ± 0.7 nM) and produced a smaller decrease in the level of activation. These results indicate that exosite 2 ligands attenuate thrombomodulin-dependent protein C activation, in part, by reducing the apparent affinity of TM456 for exosite 1 of thrombin.

DISCUSSION

Thrombin is a versatile coagulation enzyme, as exemplified by its >20 identified substrates, inhibitors, and ligands.[3,4] Consequently, thrombin activity is tightly regulated. This precision is a result of restrictive active site topology under the control of exosites that regulate accessibility and dictate

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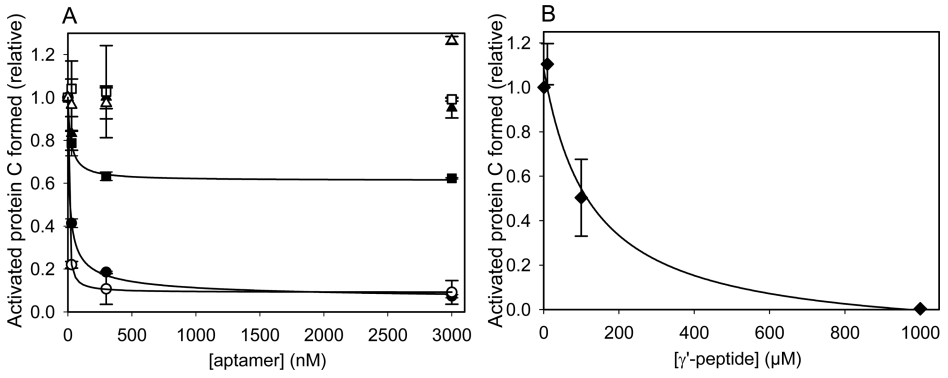


Figure 7. Effect of HD22, HD23, HD1, and γ′-peptide on thrombin-mediated protein C activation on the surface of HUVEC. Cells were grown to 80−90% confluence in 24-well plates. Washed cells were incubated with 1 μM protein C, 10 nM thrombin (filled symbols), or 20 nM RA-thrombin (empty symbols) and (A) 0−3 μM HD1 (circles), HD22 (squares), or HD23 (triangles) or (B) 0−1000 μM γ′-peptide for 30 min at 25 °C. Supernatants were removed and transferred to wells of a 96-well plate containing 100 nM hirudin and 1 mM S-2366. Activated protein C generation was quantified by monitoring the absorbance at 405 nm, and rates of substrate hydrolysis were converted to activated protein C concentration, normalized to that determined in the absence of added ligand, and plotted vs ligand concentration. Data points represent means ± SD of two determinations, and the curved lines are nonlinear regression fits of a rectangular hyperbola used to determine EC_{50}.

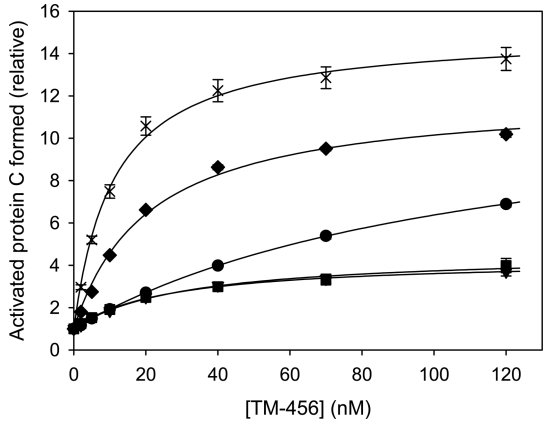


Figure 8. Effect of exosite 2 ligands on the affinity of thrombin for TM456. Protein C (200 nM) was activated by 20 nM thrombin in the presence of 0−120 nM TM456, 0.1 mM $CaCl_2$ and buffer (×), 600 nM HD1 (●), 100 nM HD22 (■), 80 μM γ′-peptide (▼), or 30 μM GpIbα269PPP (◆). Rates of activation were determined by a chromogenic assay for activated protein C and normalized with respect to that determined in the absence of TM456. Data were analyzed by nonlinear regression of a rectangular hyperbola to determine $K_{d,app}$ values (lines). Symbols represent the means ± SD of three experiments.

Table 3. Affinities of α-Thrombin for TM456 in the Presence of Exosite Ligands[a]

ligand	$K_{d,app}$ (nM)
none	11 ± 0.9
HD1	117 ± 25.8[b]
HD22	22.3 ± 3.7[b]
γ′-peptide	26.0 ± 1.6[b]
GpIbα269PPP	20.8 ± 0.7[b]

[a]Protein C activation by thrombin was assessed in the presence of increasing concentrations of TM456 and in the absence or presence of 600 nM HD1, 100 nM HD22, 80 μM γ′-peptide, or 30 μM GpIbα269PPP. Plots of activation rates vs TM456 concentration were analyzed by nonlinear regression to determine $K_{d,app}$. Values represent the means ± SD of three determinations. [b]$p < 0.05$ vs no ligand by a Student's t test.

location. The exosites are often viewed in isolation, with exosite 1 serving as the substrate docking site and exosite 2 functioning to tether thrombin to cells or effectors. However, there is mounting evidence that the exosites operate in a cooperative fashion. For example, the interaction of thrombin with γ′-fibrin(ogen) involves both exosites, and exosite 2-mediated binding of thrombin to GpIbα on the platelet surface facilitates the exosite 1-mediated interaction of thrombin with PARs. This prompts the questions of whether the exosites influence each other and whether variations in these interactions serve as another layer of regulation of thrombin's specificity. One example of this regulation is the exosite−active site allostery that alters thrombin activity.[11−14,47] Furthermore, the reverse is true because active site ligands can modulate exosite function.[16,33] Another example is reciprocal interaction between the exosites. Previously, we showed that exosite 2 occupation attenuated exosite 1-mediated binding of thrombin to fibrin(ogen).[11] Because by binding thrombin fibrin serves as an antithrombin, modulation of this interaction could have an impact on the thrombogenicity of clots.[48,49] To identify the functional consequences of this allostery, we examined the possibility that occupation of exosite 2 could modulate the cofactor function of thrombomodulin, which directs the anticoagulant and antifibrinolytic properties of thrombin.

We observed that exosite 2 ligands attenuated binding of thrombin to thrombomodulin. Binding experiments were performed using active thrombin to prevent alteration of exosite function by active site ligation.[8,47,50] Exosite communication was further demonstrated in functional assays of protein C activation, as observed previously for prothrombin fragment 2.[22,51] The response was observed with TM456 peptide, soluble thrombomodulin, and cell surface thrombomodulin. The exosite 2 ligands that promoted this were HD22, γ′-peptide, and a peptide analogue of GpIbα, ligands that contact different subdomains of exosite 2. Specificity for exosite 2 was demonstrated by weak or no response when RA-thrombin, a variant with a disabled exosite 2, was used in place of thrombin. Furthermore, HD23, a nonbinding aptamer, and scrambled GpIbα peptide elicited weak responses, highlighting the specificity of this response for exosite 2. Because the protein C activation assay is dependent on exosite 1-mediated binding of thrombin to thrombomodulin, these data verify that HD22

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and GpIbα do not bind exosite 1 in a nonspecific manner. Previous spectroscopic studies have observed a structural connection between exosites 1 and 2 upon thrombomodulin binding.[15,52,53] Importantly, the work presented here extends these findings by demonstrating that through this connection exosite 2 ligands modulate both the affinity of thrombin for thrombomodulin and its capacity to activate protein C. Therefore, our studies provide functional evidence that exosite 2 can modulate the integrity of exosite 1, further revealing inter-exosite allostery.

Protein C activation experiments were performed in the absence and presence of TM456. In both instances, exosite 2 ligands attenuated activation, with much greater inhibition in the presence of TM456. These results demonstrate allosteric regulation of both the active site and exosite 1 by exosite 2 occupation. The effect at the active site is consistent with other studies showing effects on active site fluorophores,[20,45,46,54] chromogenic activity,[14,20] and affinity of thrombin for melagatran, an active site inhibitor.[55] While these results might suggest that all of the effects of exosite 2 ligands could be mediated via the active site, the binding data indicate direct connection between the exosites irrespective of the occupancy status of the active site. Therefore, the findings reveal that the allosteric effects of ligand binding are experienced globally by thrombin, rather than solely by the active site.

The concept of thrombin as a globally allosteric enzyme leads to questions about the physiological role of such a regulatory framework. Thrombin is tightly regulated on many levels: activation, localization, substrate interaction, and inhibition. Most, if not all, of these processes are coordinated by exosites. Furthermore, there are numerous reactions that involve both exosites. Thus, chondroitin sulfate-containing thrombomodulin, which is variably expressed in different cells and tissues,[7] binds thrombin with an affinity higher than that of thrombomodulin lacking a chondroitin sulfate moiety because both exosites are ligated when chondroitin sulfate is present,[22,56] GpIbα promotes PAR cleavage by thrombin,[57] inactivation of thrombin by heparin-catalyzed heparin cofactor II involves both exosites,[58] and thrombin binds γ′-fibrin via both exosites, which affords thrombin with greater protection from inhibition.[9,27] Collectively, these observations illustrate some of the numerous ways in which the exosites regulate thrombin activity.

Our findings suggest that exosite 2 occupation has the potential to attenuate the anticoagulant and antifibrinolytic pathways driven by the thrombin−thrombomodulin complex. Among the physiological ligands for exosite 2, prothrombin fragment 2 and the γ′-chain of fibrinogen may not bind thrombin with sufficiently high affinity to modulate thrombin activity, although their concentrations may reach micromolar levels locally. However, GpIbα is sufficiently abundant and binds thrombin with high affinity[59] such that it could saturate exosite 2 and influence exosite 1 function. Likewise, β2-glycoprotein I is an abundant plasma protein that binds thrombin via exosite 2 with nanomolar affinity.[52] Another situation in which the interaction between exosite 1 and 2 may be significant is in prothrombin activation. Prothrombin fragment 1.2 binds prethrombin 2 with high affinity and modulates exosite 1, such that it could influence binding of factor Va.[60,61] This suggests that the connection between exosites 1 and 2 may play a role in prothrombin activation.

Structural studies using X-ray crystallography, mass spectrometry, and nuclear magnetic resonance have confirmed inter-exosite allostery,[8,15−18] as have dual ligand binding studies.[14,19−23,52] Although one group initially failed to observe direct communication between the exosites of thrombin,[24,25] they subsequently reported a reduced level of binding of hirudin$_{54-65}$ peptide to exosite 1 of prethrombin 2 in the presence of prothrombin fragment 1.2, which binds exosite 2.[60] These discrepancies could be the result of ligand-specific interactions, as demonstrated by inter-exosite communication mediated by PAR-1 peptide, but not by hirudin$_{54-65}$ peptide.[15] Thus, recent structural studies provide complementary evidence of communication between the exosites, and the study presented here demonstrates that this connection is functionally significant. Because thrombin is a globally flexible enzyme, regulation of its activity can be envisioned at several levels. Gross changes are achieved by binding cofactors such as thrombomodulin or heparin, where reactions with substrates and inhibitors are stimulated by orders of magnitude. More subtle changes in substrate specificity, localization, inhibition, or cofactor binding may occur simultaneously or in the presence of additional exosite ligands.[15] Thus, regulation of thrombin activity likely occurs at several levels under the influence of numerous effectors, and inter-exosite allostery may further modulate thrombin activity.

These findings provide additional support for the relevance of allosteric regulation of thrombin. Three decades of structural work indicate that thrombin is inherently flexible, and the bases for this are being unravelled. Occupation of the Na^+ site, active site, or exosite 1 serves to restrict thrombin's plasticity and lock it in a more restricted conformation.[47] Because of its proximity to the Na^+ site and the active site, it is not surprising that such global effects also extend to exosite 2. The capacity of exosite 2 occupation to modulate exosite 1 function not only complements the structural data but also may have physiological implications. Thus, in addition to directing strict thrombin localization, exosite 2 could modulate thrombin activity. This raises the possibility of site-specific regulation depending on which ligand is tethering thrombin. For example, exosite 2 may be engaged on platelets by GpIbα, β2-glycoprotein I, glycosaminoglycans, or γ_A/γ'-fibrinogen, and each of these could subject thrombin to a unique environment, as well as differentially modulating exosite 1 or the active site. Our observation that exosite 2 occupation modulates thrombin activity identifies exosite 2 as a potential target for new anticoagulants. In support of this concept, a synthetic sulfated lignin that binds exosite 2 not only modulated thrombin activity in vitro but also attenuated thrombosis without increasing the extent of bleeding in mouse thrombosis and bleeding models.[62] Therefore, our findings point to new avenues by which to modulate thrombin activity in thrombosis and hemostasis.

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Funding

This work was supported in part by the Canadian Institutes of Health Research (FRN 3992 and MOP136820) and the Heart and Stroke Foundation (T6357). P.Y.K. is supported by an Early Research Award from Hamilton Health Sciences. J.I.W.

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Biochemistry 2017, 56, 3119−3128

holds the Heart and Stroke Foundation J. Fraser Mustard Endowed Chair in Cardiovascular Research and the Canada Research Chair (Tier 1) in Thrombosis.

Notes

The authors declare no competing financial interest.

■ ABBREVIATIONS

γ_A-fibrinogen, γ_A/γ_A-fibrinogen; γ'-fibrinogen, γ_A/γ'-fibrinogen; γ'-peptide, Tyr-phosphorylated 20-amino acid analogue of the carboxy-terminal portion of the γ'-chain of fibrinogen; EC_{50}, effective concentration that causes a 50% change; GpIbα269PPP, Tyr-phosphorylated peptide analogue of residues 269−286 of glycoprotein Ibα; GpIbα269scr, scrambled, nonphosphorylated version of GpIbα269PPP; HBS, HEPES-buffered saline; HUVEC, human umbilical vein endothelial cells; PAR, protease-activated receptor; RA-thrombin, Arg93Ala/Arg97Ala/Arg101Ala thrombin variant; RU, response unit; SD, standard deviation; SPR, surface plasmon resonance; TBS, Tris-buffered saline; TM456, peptide containing epidermal growth factor domains 4, 5, and 6 of thrombomodulin.

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DOI: 10.1021/acs.biochem.7b00250
Biochemistry 2017, 56, 3119−3128